

Problem Description

In this NAFEMS contact benchmark, a steel cylinder is pressed into an aluminum block. The solution with a 2-D plane strain approximation of this problem from Nastran is compared to an analytical solution for a frictionless case.

Solution Number

SOL 400

Features Used

- Nonlinear static analysis.
- Advancing contact area with curved contact surfaces
- Deformable to deformable contact

Reference

1. Benchmark 1 from NAFEMS publication R0094, Advanced Finite Element Contact Benchmarks, 2006.

Modeling Techniques Used

Steel cylinder and aluminum block are modeled as two contact bodies with either linear or quadratic plane strain elements. The model details are given below:

Material Properties

$E_{cylinder}$	=	210 kN/mm ²
$\nu_{cylinder}$	=	0.3
E_{block}	=	70 kN/mm ²
ν_{block}	=	0.3

Loading

Vertical point load $F = 35\text{kN}$

Boundary Condition

Symmetric displacement constraints along vertical symmetric line

Bottom surface of the block is fixed ($U_x = U_y = 0.0$)

Touching contact between cylinder and block with segment to segment contact

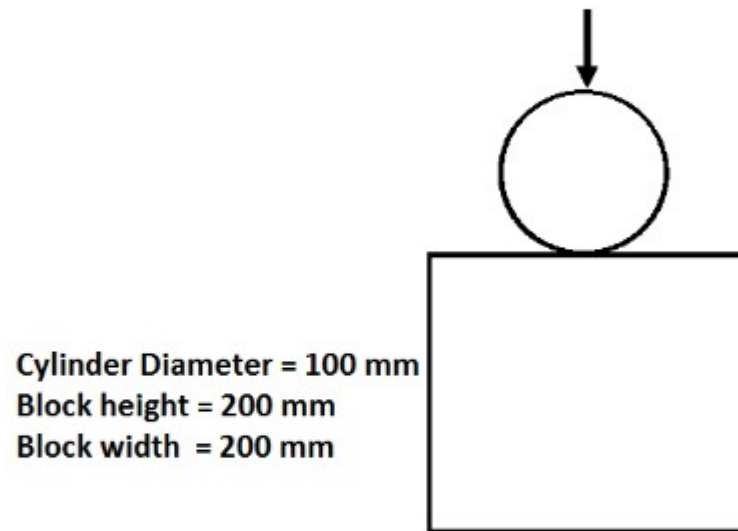
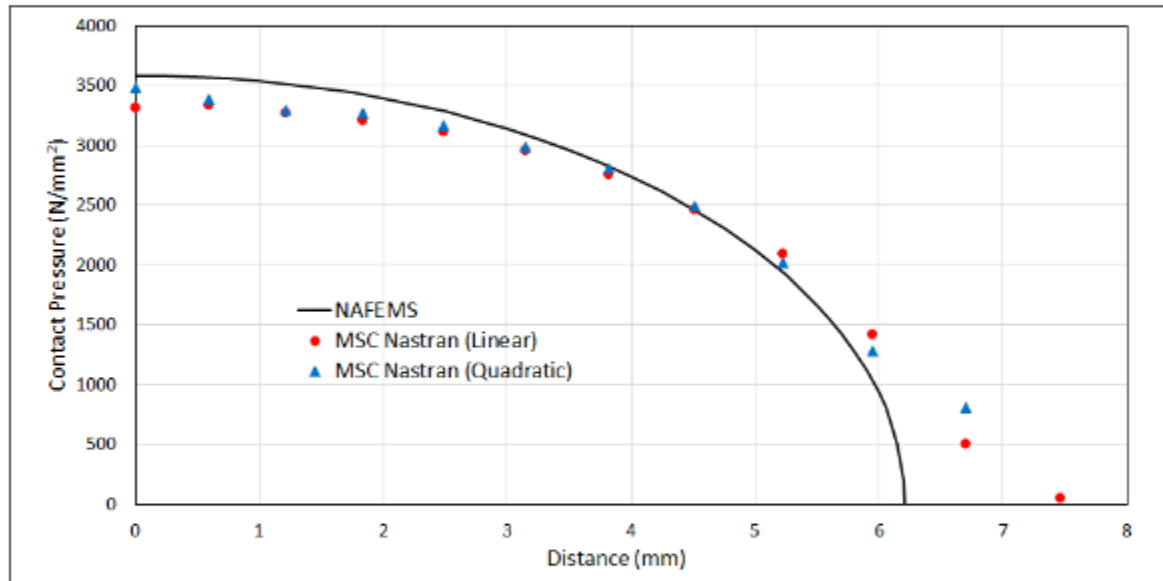
Illustrations of the Model

Figure 6.68-1 2D Cylinder Roller Contact

Model File Name and Location

Model Name	Description	Location
nafem_r00941a.dat	Linear element	\verifman\nafems
nafem_r00941b.dat	Quadratic element	\verifman\nafems

Results



Element	a_{min} (mm)	a_{avg} (mm)	a_{max} (mm)	% Error	P_{max}	% Error
Linear	6.698577	7.0811275	7.463678	14.03%	3335.70	-6.96%
Quadratic	5.952867	6.325722	6.698577	1.86%	3486.60	-2.75
NAFEMS		6.21			3585.37	

